Gemini — vision + reasoning | Groq — text fallback | Local RAG, 58 chunks | No vector database

The AI interprets. The code decides.

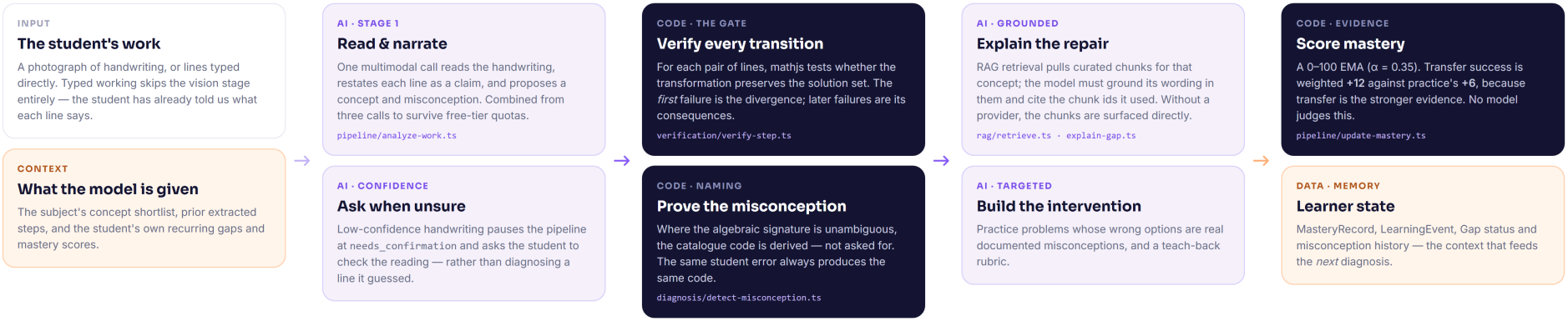
A language model is the only thing that can read handwriting or infer what a student was attempting. It is not a source of truth about whether one line follows from another. GapFinder separates those two capabilities at the architectural level, not by prompt engineering — which is why the product can accuse a specific line by name.

AI PIPELINE

The model interprets. The code decides.

Colour marks *authority*, not technology. Nothing the model says about correctness reaches the student until deterministic code has confirmed it.

AI-generated interpretation — reads, narrates, names, phrases · **Deterministic logic** — decides truth, mastery and eligibility · **Persistent learner data** — remembered across sessions



When the model is unavailable, the pipeline does not stop

A free-tier rate limit is a certainty, not an edge case. Every stage after reading has a deterministic substitute.

Classification
The offline classifier reads the same verified algebra and picks the concept from the shape of the error — reporting low confidence, out loud.

Explanation
Retrieved chunks are surfaced directly alongside the verifier's account of what changed.

Reading a photo
The one stage with no local substitute. The app says so and offers typed working, which runs the identical pipeline with no AI at all.

GapFinder

Diagram 03 — AI pipeline & authority boundaries

THE AUTHORITY BOUNDARY

Who is allowed to decide what

The test applied to every feature: *if the model returned nonsense here, would a student be told something false?* Where the answer is yes, the feature is built a different way.

AI-GENERATED INTERPRETATION

Reads, narrates, names, phrases

Reading, intent, concept slug, practice drafts, wording. Always *input* to the next stage — never a verdict.

DETERMINISTIC LOGIC

Decides truth and eligibility

Whether a step follows, where the divergence is, what the correction reads, and whether mastery may be claimed.

PERSISTENT LEARNER DATA

Remembers across sessions

Gaps, mastery EMAs, recurring codes, events, roadmap order. Written once at computation, read back thereafter.

DECISION	OWNER	WHY IT SITS THERE
What do the marks say, and what was the student attempting?	GEMINI VISION	Only a model can read handwriting or infer intent from an incomplete attempt. Low confidence pauses the pipeline and asks the student rather than guessing.
Does step *n* follow from *n*−1?	MATHJS	Must be provable. A model asked "is this step right?" is confidently wrong often enough to be useless for an accusation.
Where is the first divergence, and which later lines are its consequences?	SOLUTION-AUDIT.TS	The product's core claim, computed and measured by the eval harness. A line faithful to a wrong premise is not a new mistake.
What should that step have read?	SOLVE-STEP.TS	Derived from the student's own opening line. Never generated — a generated "correct" line could be wrong.
Which misconception is it?	SIGNATURE or CATALOGUE PICK	An unambiguous algebraic signature **proves** the code. Otherwise a model picks from the catalogue and it is labelled **matched**.
Which concept broke?	GEMINI → structural fallback	Needs judgement. With no provider, the offline classifier picks from the shape of the verified error.
Is a generated practice problem valid?	SOLVED INDEPENDENTLY	An unverified problem is never displayed.
Has the student mastered this?	EXAM/VERDICT.TS	Rules, not opinion: two questions minimum, sound reasoning required, a returning code decisive.

INSIDE ONE ANALYSIS

What the model receives, and what it returns

Stage boundaries are written to `analysis.status` as they are entered. The Analyzing screen polls that, so the progress a student watches is the real pipeline position rather than an animation on a timer.

WHAT IS SENT

The prompt payload

The work	A client-side downscaled and compressed photograph, or the student's typed lines
Subject	Chosen at capture; narrows the concept shortlist
Concept shortlist	The seeded slugs for that subject, so the model selects rather than invents
Learner context	Recurring misconception codes and mastery scores for the concepts in play
Output schema	A Zod schema reduced to the subset Gemini's structured-output endpoint accepts

WHAT COMES BACK

Structured, never prose to be parsed

Extracted steps	Each line as read, with a per-line confidence and a `needsConfirm` flag
Reasoning claims	Each line restated as what the student was asserting
Proposed concept	One slug from the shortlist — checked against the catalogue before use
Proposed code	A misconception code, discarded if it is not in the catalogue
Explanation	Grounded in retrieved chunks, with the chunk ids it used recorded

Three calls became one. Reading, narrating and classifying were once separate requests — three round trips and three charges against a free-tier quota for a single photograph. They are now one multimodal call whose result is carried forward, so later stages never re-ask what they were just told. This is only safe *because* nothing the model says about correctness is trusted downstream.

GROUNDING & PERSONALISATION

Where the words come from, and why they are about *this* student

Two separate mechanisms, often conflated. Retrieval decides what is *said*; the learner record decides what is *selected* and how the next call is framed.

01 · RETRIEVAL WITHOUT A VECTOR DATABASE

58 curated chunks, scored locally

Explanations are grounded in `KnowledgeChunk` rows retrieved per concept and filtered by kind — `explanation`, `misconception`, `teaching_strategy`, `worked_example`, `practice_pattern`. Scoring is local TF-IDF-style keyword overlap. No embedding API, no vector store, no recurring cost — and a corpus small enough for a human to read end to end.

Asked by name	Asked about by name with nothing scoring, **all** of that concept's chunks are returned — everything filed under it is relevant by construction
Cited for an error	If nothing scores, **nothing** is cited. An unrelated chunk is worse than none

02 · RESEARCH SOURCES

A model is never asked for a citation

Titles, authors, DOIs and URLs come back from provider APIs — **OpenAlex**, **Crossref** and **arXiv**, all free and keyless — and carry their provenance. The research layer contains no prompt at all. A fabricated citation that looks correct is worse than no citation, because a student would cite it.

Query mapping	Each concept slug maps to terms the literature actually uses; a misconception's display name is prose and matches on filler words
Two gates	A result must carry a **distinctive** on-concept anchor *and* an education signal, or it is discarded
Silence is named	Timed out, nothing relevant, no anchors, subject unsupported — four messages, never a blank panel

03 · PERSONALISATION

What makes the next analysis different from the first

Into the prompt. Recurring misconception codes and mastery scores for the concepts in play are passed as context, so the model chooses against a background rather than from scratch.

Into the selection. Which intervention is offered — more practice, transfer, teach-back — is decided in code from the gap's status and the mastery record, not by asking a model what would be nice.

Into the verdict. Exam Mode is given the codes this student held *before* the exam. A resurfacing code is decisive against mastery whatever the score.

FAILURE HANDLING

The pipeline keeps working when its dependencies do not

Every stage after reading a photograph has a deterministic substitute, so a provider outage degrades confidence rather than breaking the product — and the interface says which it is.

THE CASCADE

cache → Gemini → Groq → deterministic

Gemini stays first: its vision is the strongest available here and the prompts are tuned to it. Groq's free tier is separate from Google's, so a Gemini quota exhaustion does not stop the product working.

FAILURE	BEHAVIOUR
`quota`	**No retry** — a free-tier limit is not transient
`no_key`	No retry. Skip to the next provider
`unsupported`	No retry — e.g. vision from a text-only provider
`network`	Retry in-provider, backoff 300 ms × attempt
`invalid_response`	Retry — schema validation may be transient

THE FAILURE THE PRODUCT REFUSES

"Couldn't check" is never reported as "nothing wrong". If every line came back `uncertain`, the analysis fails honestly. If only some did, the result says so: *"2 lines couldn't be checked, but everything we could verify holds."*

WHEN BOTH PROVIDERS ARE GONE

Classification	The offline classifier reads the same verified algebra and picks the concept from the shape of the error — reporting **low** confidence, out loud
Explanation	Retrieved chunks are surfaced directly alongside the verifier's account of what changed
Practice	Generated from templates, each solved independently before display
Verification	Untouched. It never used a model in the first place

THE ONE STAGE WITH NO SUBSTITUTE

Reading a photograph. Nothing local turns pixels into equations. The failure returns a `TYPE_INSTEAD:` prefix and the app offers typed working, which runs the *identical* pipeline with no AI involved at all.

FAILURES NAME THE PATH THAT STILL WORKS

`QUESTION_ONLY:`	A photographed question with no attempt — routed to guided solving, not a dead end
`TYPE_INSTEAD:`	Image reading failed — typing gets the same diagnosis

OBSERVABILITY & EFFICIENCY

What actually ran, and what it cost

Every model call is recorded, so the pipeline can always account for itself — which provider served each stage, how long it took, and what it retrieved.

AI OBSERVABILITY · `/dev/observability`

Every model call is inspectable after the fact

`AiUsageLog` records, per call: the pipeline stage, `provider:model`, whether it was served from cache, whether it succeeded, latency, the error text when it did not, and the RAG chunk ids that fed the prompt. A per-analysis trace view replays the sequence.

This makes two things visible that are usually invisible: a **failover** — Gemini declined, Groq answered — and a **fallback**, where no provider answered and deterministic logic produced the result. Retrieval becomes traceable rather than assumed.

QUOTA MANAGEMENT

Free tiers are the design constraint

Response cache	`AiCallCache`, keyed by a content hash, 7-day TTL. A hit is logged as its own provider
Small keys	The key hashes the prompt, the system instruction and a **digest** of the image — not the image bytes
Request collapsing	Three pipeline calls became one multimodal call
Structured output	JSON schema on both providers — no reparse loops
Resource cache	14 days; a concept's literature changes monthly, not hourly
No regeneration	Opening a page again never re-runs an analysis

WHAT THE EFFICIENCY WORK BOUGHT

Round trips	Four sequential calls collapsed into **one** multimodal call, carried forward so no stage re-asks
Bytes on the wire	Photographs are downscaled and compressed **in the browser** before transmission
Perceived latency	`waitUntil` returns the response before background persistence finishes; resources load after the diagnosis renders
Repeat cost	Content-hashed caching, and a stored analysis is never regenerated — a revisited page costs nothing